MACHINE LEARNING • MINI PROJECT

Latent Feature Extraction for Musical Genres from Raw Audio



Reimplementation, critical evaluation, and loss-weight sensitivity analysis

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GTZAN dataset | Classical · Jazz · Metal · Pop

The question behind the project

Can a neural network learn useful genre representations directly from raw audio—without spectrograms or hand-engineered audio features?

- Reimplement the Deep Softmax Autoencoder (DSAE) proposed by Sawhney et al. (2018).
- Compare it with a simpler two-layer neural network and a loudness-only classifier.
- Test whether results survive song-level splitting and amplitude normalization.
- Study how the reconstruction-loss weight γ affects classification.

Core concern

High accuracy may reflect recording loudness or song identity—not generalizable musical timbre.

Evaluation principle

Use a clean song-level split so clips from the same song cannot appear in both training and test sets.

Four target genres

Classical • Jazz • Metal • Pop

Dataset & raw-audio pipeline

399	30 s	22,050 Hz	11,968	500-D
songs	per original track	mono audio	one-second clips	clip vector

01 30-second song
GTZAN

02 Split into 1-second clips
22,050 samples

03 Average pool by factor 44
22,000 → 500

04 Standardize
Train mean & SD

No Fourier transform, filterbank, or spectrogram is computed. The model receives a compact representation of the waveform itself.

Why the split strategy matters

Random clip-level split • original paper

Clips from the same song can land in both train and test sets. The model may recognize recording/song identity.

Song-level split • this project

All clips from a song stay in one partition. This better tests generalization to unseen songs.

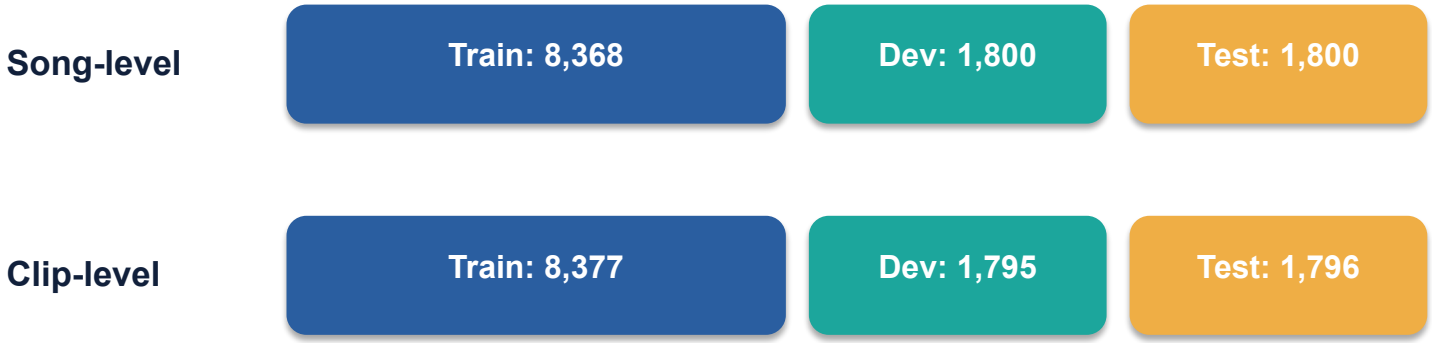
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The clean song-level split prevents song leakage; the random clip split does not.

Models evaluated

1 Loudness-only baseline

Feature: log(std(clip) + 1e−8)

A logistic regression model uses just one scalar: clip loudness.

Purpose: check whether volume alone predicts genre.

2 Two-layer neural network

500 inputs → 128 hidden units (tanh) → 4-class softmax

Cross-entropy objective.

Purpose: reproduce the simpler neural baseline.

3 Deep Softmax Autoencoder

Encoder: 500 → 256 → 192 → 128 → 64

Decoder: 64 → 128 → 192 → 256 → 500

Classifier: 64 → 32 → 16 → 4

Learns a 64-dimensional bottleneck.

The DSAE jointly reconstructs the input waveform and predicts its genre.

How the Deep Softmax Autoencoder learns

Joint loss function

$$L = \gamma \times \mathrm{MSE}(\text{reconstruction}, x) + (1 - \gamma) \times \mathrm{CrossEntropy}(\text{logits}, y)$$

Reconstruction loss • MSE

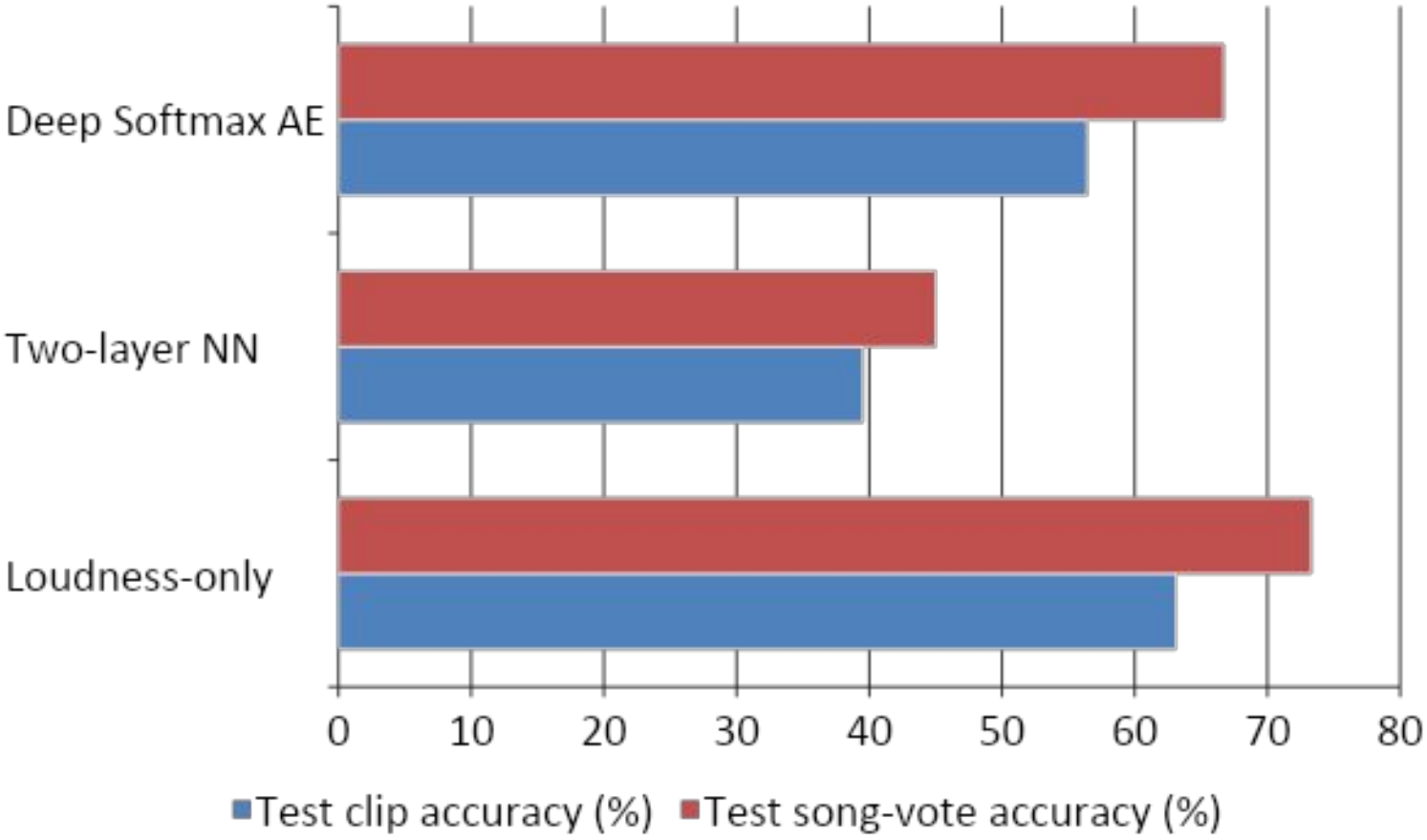
Encourages the 64-D bottleneck to preserve information about the input waveform.

Classification loss • Cross-entropy

Uses genre labels to make the bottleneck useful for four-way genre prediction.

Paper default: $\gamma = 0.90$ → 90% reconstruction loss + 10% classification loss

Results on the clean song-level split



Best clip accuracy

63.1%

Loudness-only model

Best song-vote accuracy

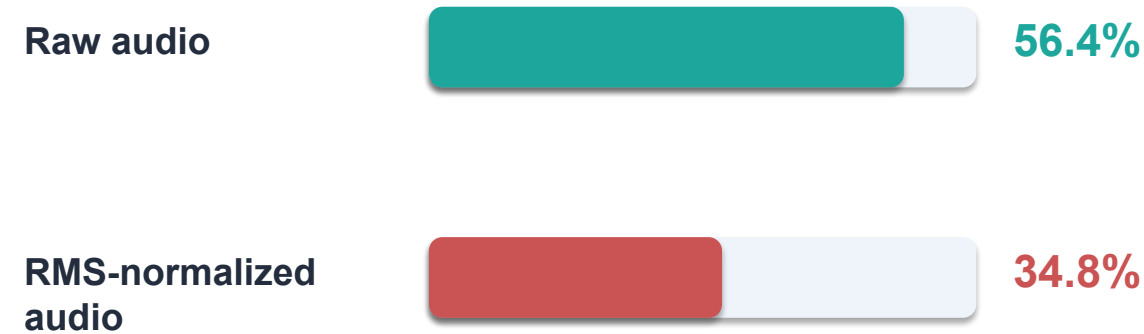
73.3%

Loudness-only model

Important: a one-feature model beats the complex network.

The loudness-bias test

DSAE test clip accuracy



What changed?

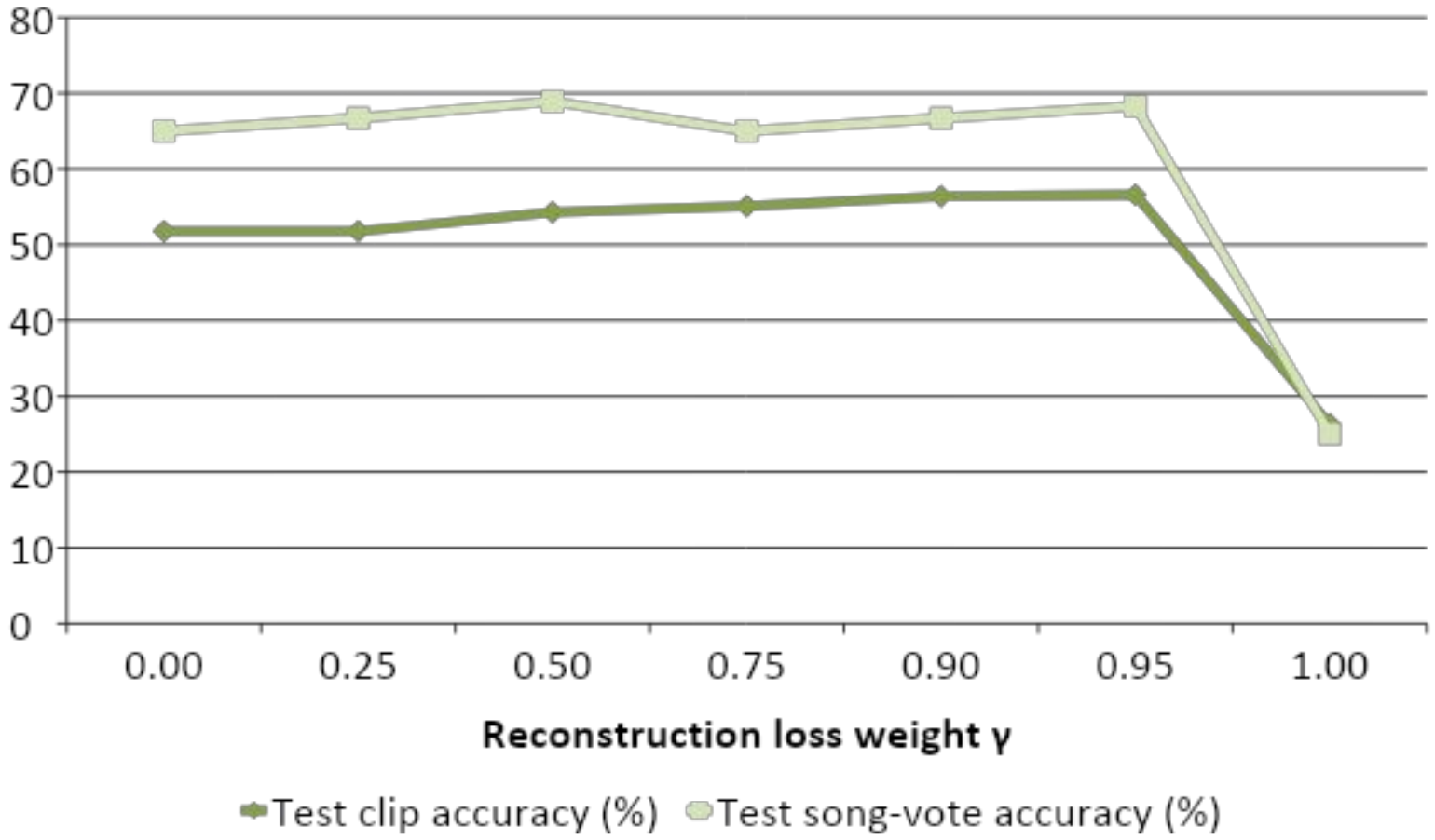
Each clip was RMS-normalized so every clip had approximately equal energy.

What happened?

Accuracy fell from 56.4% to 34.8%—only modestly above 25% chance for four classes.

Interpretation: the raw-audio model relies heavily on amplitude cues; average pooling may also discard fine pitch and harmonic detail.

Extension: sensitivity to loss weight γ



Best song-vote result

68.9% at γ = 0.50

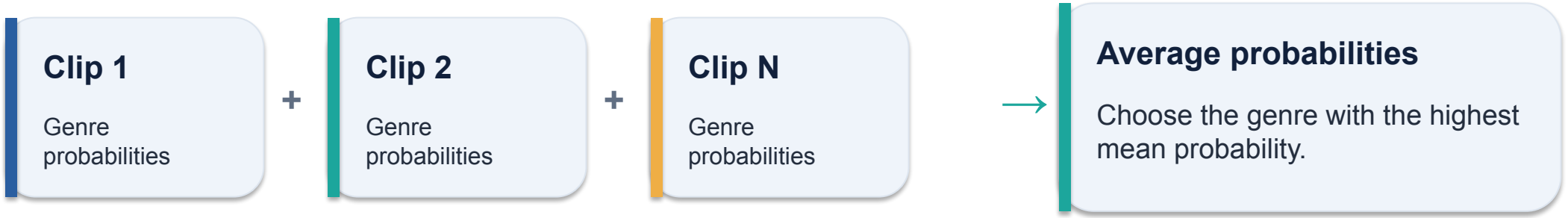
Pure autoencoding

25.0% song-vote accuracy at γ = 1.00

Reconstruction helps—but supervision is essential.

Why predict at the song level?

Individual 1-second clips can be noisy. We average their softmax probabilities to make one prediction per song.



Observed song-vote improvement over clip accuracy

+5.5 pp
Two-layer NN

+10.3 pp
Deep Softmax AE

+10.2 pp
Loudness-only

What the experiments tell us

1 Loudness is a strong confounder

A single loudness feature reaches 63.1% clip and 73.3% song-vote accuracy.

2 Clean splits matter

Song-level splitting is more credible for evaluating performance on unseen recordings.

3 Joint learning can regularize

γ between 0.50 and 0.95 performs relatively well; γ = 1.00 collapses to chance.

4 Raw waveform pooling is limited

Compressing 22,000 samples to 500 values can remove pitch, harmonic, and transient detail.

Main takeaway: strong benchmark numbers do not automatically prove that a model learned meaningful musical timbre.

Conclusion & next steps

- Reimplemented the DSAE on four GTZAN genres using a song-level split.
- DSAE achieved 56.4% test clip accuracy and 66.7% song-vote accuracy on raw audio.
- Loudness-only classification outperformed the deep model, exposing a major recording-level bias.
- The gamma ablation showed a useful regularization range, while pure autoencoding reached chance.

Future work

Compare against richer representations such as spectrograms or Mel features; evaluate on more songs and datasets; test stronger normalization and augmentation.

References

Sawhney, Vasavada & Wang (2018), “Latent Feature Extraction for Musical Genres from Raw Audio.”

Tzanetakis & Cook (2002); Sturm (2014).

Thank you

Questions?